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Imidacloprid disrupts larval molting regulation and nutrient energy metabolism, causing developmental delay in honey bee *Apis mellifera*

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Abstract

Imidacloprid is a global health threat that severely poisons the economically and ecologically important honeybee pollinators, *Apis mellifera*. However, its effects on developing bee larvae remain largely unknown. Our pilot study found that imidacloprid causes developmental delay in bee larvae, but its toxicological mechanisms are not yet fully understood. In this study, we exposed bee larvae to imidacloprid at environmentally relevant concentrations of 0.7, 1.2, 3.1, and 377 ppb. A significant dose-dependent delay in the larval development was observed, including reductions in body mass, width, and growth index; however, imidacloprid did not affect larval survival and food consumption. Predominant toxicological effects induced by high concentrations of imidacloprid (377 ppb) included disruption of neural transmission, induction of oxidative stress and gut structural damage and apoptosis, inhibition of developmental regulatory hormones and genes, disruption of proteolysis, amino acid transport, protein synthesis, carbohydrate catabolism, oxidative phosphorylation, and glycolysis energy production. In parallel, we observed that the larvae employed antioxidant defense and P450 detoxification mechanisms to counteract imidacloprid. Ultimately, this study provides the first evidence that environmental exposure to imidacloprid can retard bee larval growth and development by disrupting molting regulation and limiting the metabolism and utilization of dietary nutrients and energy. These findings have broader reference implications for studies assessing pesticide hazards to other juvenile animals.

eLife assessment

This is a **useful** investigation of the changes in gene expression and some of the physiological consequences of sublethal exposures to the neonicotinoid insecticide imidacloprid in honeybee larvae. While the study adds to our understanding of how this insecticide impacts development and growth of honeybees, the evidence supporting the major claims is **incomplete**. The study would benefit from a more thorough mechanistic characterization of the phenotypes to substantiate the conclusions.

1. Introduction

Hazardous pesticides are extensively used worldwide to safeguard crop production, particularly in developing countries that depend on agriculture. However, exposure to these chemicals has become a global health concern. Imidacloprid, introduced in 1991, is known for its high animal toxicity and has received international attention (Jeschke et al., 2011). The World Health Organization and the U.S. Environmental Protection Agency have listed it as an acute oral and dermal toxicant for animals. Overuse of imidacloprid can lead to environmental contamination, endangering human and animal health through food chains, poisoning non-target pollinators, and disrupting ecological balances (Tudi et al., 2021). Adults exposed to high doses of imidacloprid experience dizziness, apathy, movement disorders, labored breathing, and temporary growth retardation. Imidacloprid also causes adverse congenital, hematological, hepatic, and renal effects, along with degenerative changes in various organs (Khan et al., 2010; Shaw et al., 2014; Yang et al., 2014). In rats and rabbits, imidacloprid causes severe liver, thyroid, and body weight problems, reproductive toxicity, developmental retardation, and neurobehavioral deficits, even at low doses (Anatra-Cordone M, 2005). Imidacloprid also affects the singing ability of birds, making it difficult for them to attract mates and reproduce. Imidacloprid represents a significant contributing factor to the decline in the number of terrestrial insects since 1925, which has declined by an average of 9% per decade (Cardoso et al., 2020; Sánchez-Bayo and Wyckhuys, 2019).

The development and health status of juvenile animals play a critical role in population structure and future growth (Gill et al., 2012), but they are more susceptible to the toxic effects of pesticides than adults. However, there is a paucity of information on the assessment of toxic effects of imidacloprid on juvenile animals. Available reports primarily describe the epistatic traits influenced by toxicity, with limited investigation into the biochemical mechanisms involved. Studies have shown that imidacloprid poses a potential health risk to children through dietary exposure (Anderson and Harmon Threatt, 2021) and can lead to developmental abnormalities in amphibians (Samojeden et al., 2022) and fish (Islam et al., 2019) as well as negative effects on insect pupal development (Jia and Li, 2023). In insects, the honeybees (*Apis mellifera* L.) are valuable model organisms with significant economic and ecological importance as pollinators of crops and maintenance of ecological balance (Zheng and Fu-Liang, 2009). However, the widespread use of pesticides, including imidacloprid, has resulted in colony collapse disorder (CCD), causing serious impacts on beekeeping industry and ecological balance (Mahé et al., 2021). Studies have shown that Imidacloprid is applied as seed coating (Simon-Delso et al., 2015) and accumulate in the plant (Dively and Kamel, 2012; Rocha et al., 2022), exposing non-target bees to imidacloprid residues during pollen and nectar collection (Tong et al., 2018). Even low levels of systemic imidacloprid cause sublethal effects in adult bees, including neurotoxicity (Belzunces et al.,

2012), behavioral changes (Karahan et al., 2015; Medrzycki et al., 2003), reproductive impairment (Chaimanee et al., 2016), immunity (Pettis et al., 2013), apoptosis and autophagy (Carneiro et al., 2022), and shortened lifespan (Anderson and Harmon-Threatt, 2021). Larvae is the future of colony growth and development, yet the broader toxic effects of imidacloprid on larvae remain unknown. Limited reports suggest that developing bee larvae are exposed to imidacloprid through contact with contaminated hives or ingesting contaminated food (Bhame et al., 2018), leading to delayed development, reduced rates of capping, pupation, and eclosion (Yang et al., 2012), and impaired olfactory learning behavior (Peng and Yang, 2016). In the pilot study, we found that imidacloprid exposure caused obvious developmental delay in bee larvae, but the molecular and biochemistry mechanisms remain to systematically investigated. To this end, this study comprehensively evaluated the toxicity of imidacloprid on honey bee larvae at multiple levels including histology, neurotoxicity, molting regulation, oxidative stress, detoxification, nutrition, and energy metabolism, and provides first-hand evidence for imidacloprid causing developmental delay in bee larvae. Because the mechanisms involved are relatively conserved in animals, our findings have broader implications for understanding and assessing the potential risk of continuous exposure to imidacloprid during animal development.

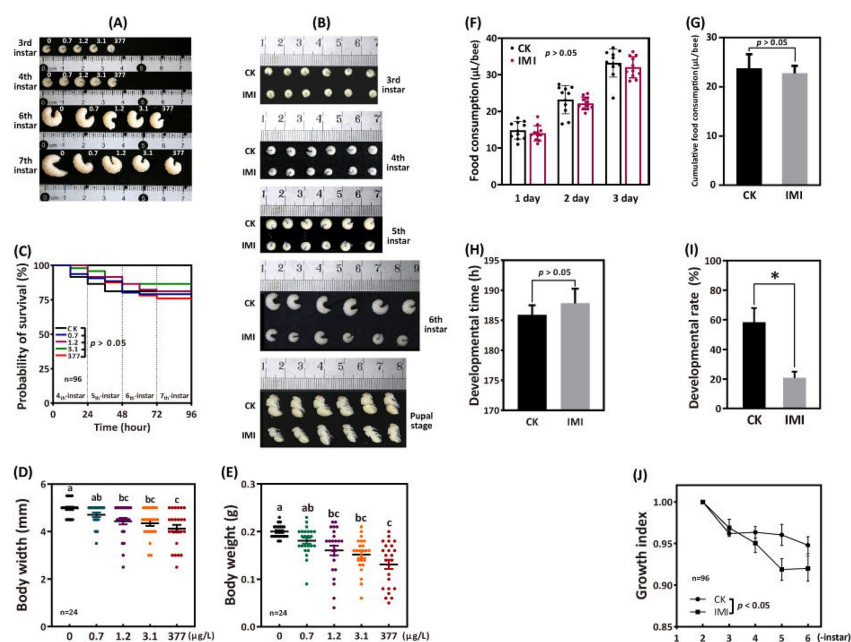
Results

Imidacloprid causes bee larval developmental retardation

During the 96-hour observation period, larval survival in the four imidacloprid-exposed groups was not significantly different from that in the control group. However, with increasing exposure concentration and duration, larval developmental progress, weight and width were significantly reduced (Fig. 1A–1E). Further analysis focused on the most adverse exposure level of 377 ppb, which did not affect survival, feeding (Fig. 1C, 1F, and 1G), and the time taken for larvae to successfully enter the pupal stage (Fig. 1H), but resulted in slow developmental obstacles and lower initial rate of successful pupation and growth index. In addition, imidacloprid also caused consistently lower weight (Fig. 1I) and width (Fig. 1J) of larvae up to the pupal stage (Fig. 1B).

Figure 1.

Imidacloprid causes larval developmental retardation in *Apis mellifera*. Larvae were orally administered imidacloprid (IMI) or artificial food (CK) beginning at three-day-old, followed by developmental monitoring for 96 hours. (A) Larval developmental phenotypes at 0.7, 1.2, 3.1, and 377 ppb environmental concentrations of imidacloprid exposure; (B) Larval developmental phenotype at 377 ppb imidacloprid exposure; (C) Effect of 0.7, 1.2, 3.1, and 377 ppb imidacloprid on larval survival; (D) and (E) body width and weight of larvae exposed to 0.7, 1.2, 3.1, and 377 ppb imidacloprid for 96 hours; (F) and (G) statistics for daily (F) and three-day cumulative feeding (G)



for larvae exposed to 377 ppb imidacloprid; (H) larval development time from three-day-old to pupal stage; (I) The number of larvae that successfully reached the pupal stage as a percentage of the total initial sample size; (J) The growth index of larvae. Statistical significance was set at $*P < 0.05$ and $**P < 0.01$.

Imidacloprid neurotoxicity in bee larvae

Under imidacloprid environmental stress at 377 ppb, AChE activity, which is closely related to the disruption of neurotransmission, was significantly inhibited in larvae (Fig. 2A). Neither acetylcholine receptor alpha 1 (*Alpha1*) nor acetylcholinesterase 2 (*Ace2*) gene expression was significantly different from the control. In contrast, acetylcholine receptor alpha 2 (*Alpha2*) and acetylcholinesterase 1 (*Ace1*) gene expression was significantly higher than that in the control group (Fig. 2B).

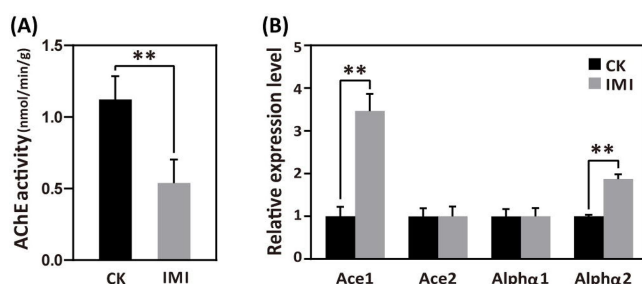


Figure 2.

Imidacloprid neurotoxicity in *Apis mellifera* larvae. (A) AChE activity and (B) nerve conduction-related gene expression analysis in larvae exposed to 377 ppb imidacloprid for 72 hours or a control group. IMI is the larvae exposed to imidacloprid for 72 hours. CK is the artificial food control group. Statistical significance was set at $*P < 0.05$ and $**P < 0.01$.

Imidacloprid disrupts the homeostasis of developmental regulation in larvae

In imidacloprid-exposed larvae, the developmental regulatory gene juvenile hormone acid methyl transferase (*JHAMT*) and the transcription factor broad complex (*Br-c*) were downregulated, while vitellogenin (*Vg*) gene expression was upregulated by seven-fold in comparison to control groups (Fig. 3A). In parallel, imidacloprid exposure decreased the major regulator of insect metamorphosis 20-hydroxyecdysone (20E) titers but had no effect on preventing premature metamorphosis juvenile hormone (JH-3) titers (Fig. 3B).

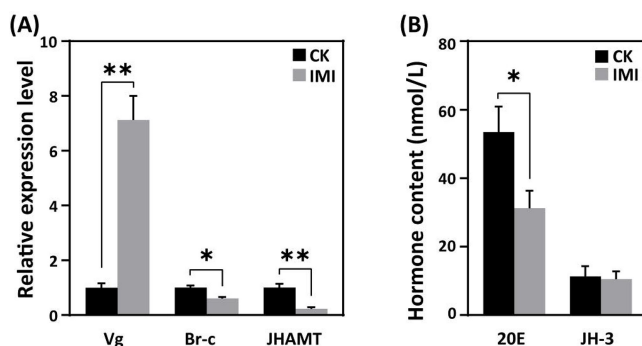


Figure 3.

Effects of imidacloprid on the homeostasis of developmental regulation in *Apis mellifera* larvae. (A) Relative gene expression and (B) hormone levels of developmental regulatory-related genes in larvae exposed to 377 ppb imidacloprid for 72 h or a control group. Statistical significance was set at * $P < 0.05$ and ** $P < 0.01$.

Imidacloprid induces detoxification in larvae

According to the qRT-PCR analysis, the expression levels of cytochrome P450 monooxygenase family members such as CYPq1, CYPq2, CYP450, CYP6AS14, CYP4G11, and CYP306A1, except for CYPq3, were observed to be significantly upregulated by several fold in imidacloprid-induced developmental retardation larvae. Among them, CYPq1 showed an abnormal 94-fold increase (Fig. 4).

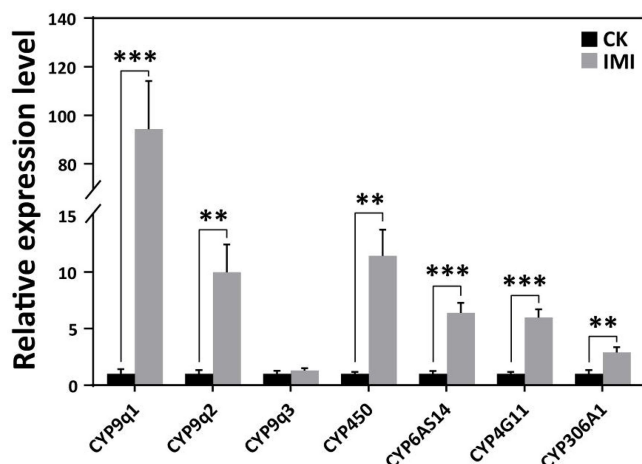


Figure 4.

Imidacloprid induced a detoxification response in *Apis mellifera* larvae. Relative expression levels of cytochrome P450 monooxygenase family genes in larvae exposed to imidacloprid for 72 hours. IMI is the larvae exposed to imidacloprid for 72 hours. CK is the artificial food control group. Statistical significance was set at * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

Imidacloprid toxicity causes oxidative stress and induces antioxidant defense in larvae

Larvae exposed to imidacloprid exhibited a significant increase in reactive oxygen species (ROS) levels (Fig. 5A) and upregulation of the antioxidant genes GPX and Trx expression (Fig. 5C), as well as increased activities of the antioxidants CAT, SOD and GSH (Fig. 5B), resulting in a significant increase in total antioxidant capacity (T-AOC) (Fig. 5B) compared to the control group. Under these conditions, MDA levels, used as a marker of lipid peroxidation,

were observed to increase, while protein carbonylation damage indicator PCO levels remained unchanged (Fig. 5A).

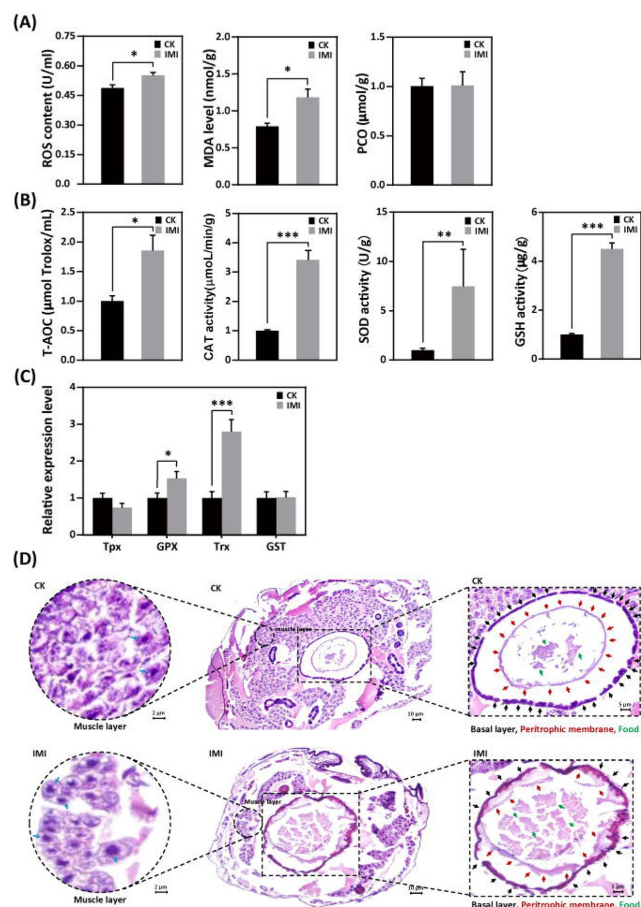


Figure 5.

Imidacloprid exposure causes oxidative stress, gut apoptosis, and tissue structural damage while inducing antioxidant defenses in *Apis mellifera* larvae. (a) Analysis of oxidative stress and damage in larvae exposed to imidacloprid for 72 hours. ROS levels indicate the degree of oxidative stress. MDA is a marker for lipid peroxidation. PCO levels reflect the extent of protein carbonylation damage; (b) Activity of antioxidants in larvae exposed to imidacloprid for 72 hours; (c) The relative expression levels of antioxidant genes in larvae exposed to imidacloprid for 72 hours; (d) Histological sections of the larval gut stained with H&E. The circled inserts in the figure show magnified views of the muscle layer. The blue arrow indicates the cell nucleus of the muscle layer. The boxed inset in the figure is a magnified view of the basal layer (black arrow), peritrophic membrane (red arrow), and food residues (green arrow). IMI is the larvae exposed to imidacloprid for 72 hours. CK is the artificial food control group. Statistical significance was set at * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

Imidacloprid induces gut apoptosis and tissue damage in larvae

Hematoxylin-eosin staining revealed a significant reduction in the number of cells in the muscular layer of the larval gut exposed to imidacloprid with clear apoptotic characteristics compared to the standard feeding control groups (Fig. 5D). This was evidenced by increased nuclear staining and compacted chromatin, as well as disrupted cell arrangement in the basal layer of the gut, increased nuclear staining of basal lamina cells, and apoptotic signals. The integrity of the peritrophic membrane structure of the gut was better in the control group, while imidacloprid-exposed larvae were still in the formation stage, resulting in the accumulation of undigested food residues in their guts (Fig. 5D).

Imidacloprid impairs larval digestion and breakdown of dietary nutrients

Exposure to imidacloprid did not significantly affect the daily food intake of developmentally delayed larvae compared to controls (Fig. 1F and 1G). However, imidacloprid toxicity suppressed the expression of nutrient catabolism genes, including alpha-amylase (α -Am) and alpha-glucosidase (α -Glu), which are involved in carbohydrate catabolism (Fig. 6A), carboxypeptidase (CPs) and aminopeptidase (APs) involved in

proteolysis (Fig. 6B), and glycine tRNA ligase (*Aats-Gly*) and tyrosine tRNA ligase (*Aats-Tyr*) involved in amino acid transport (Fig. 6C).

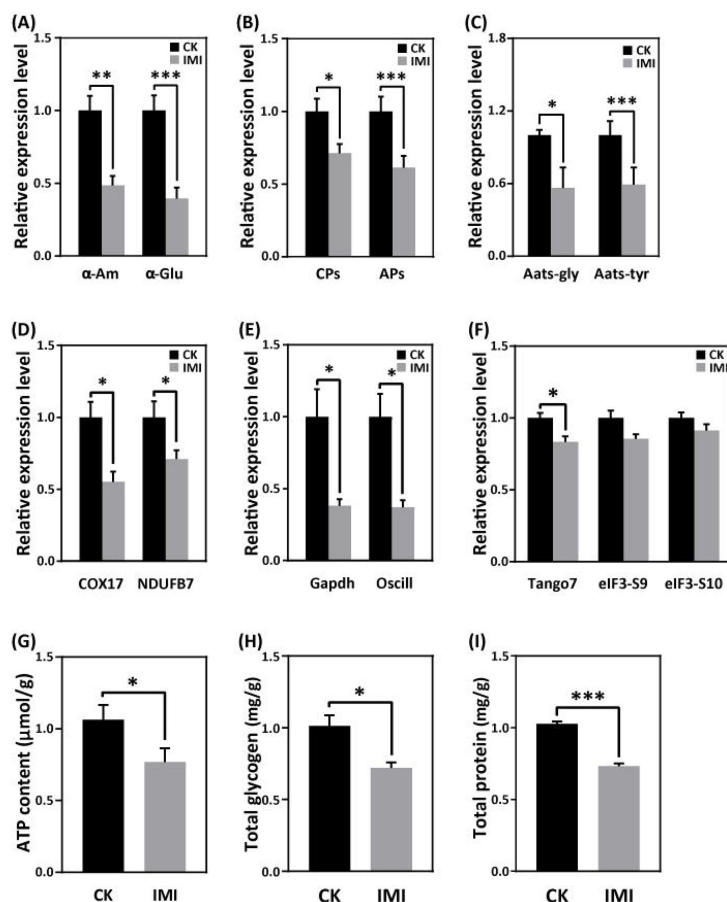


Figure 6.

Imidacloprid toxicity inhibited the expression of genes involved in nutrient catabolism, protein synthesis, and energy metabolism and reserves. (A) Carbohydrate catabolism; (B) Proteolysis; (C) Amino acid transport; (D) Mitochondrial oxidative phosphorylation; (E) Glycolysis; (F) Protein synthesis. (G) The total contents of ATP, glycogen (H), and protein (I). IMI is the larvae exposed to imidacloprid for 72 hours. CK is the artificial food control group. Statistical significance was set at * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

Imidacloprid causes restriction of larval protein synthesis and energy metabolic dysfunction

RT-qPCR analysis revealed that imidacloprid exposure reduced the expression of critical genes involved in energy metabolism, including cytochrome c oxidase copper chaperone (*COX17*) and NADH dehydrogenase [ubiquinone] 1 β subcomplex subunit 7 (*NDUFB7*), in developmentally delayed larvae (Fig. 6D). Imidacloprid exposure also inhibited the expression of genes related to glycolysis and energy production, such as glyceraldehyde-3-phosphate dehydrogenase (*Gapdh*) and glucosamine-6-phosphate isomerase (*Oscillin*) (Fig. 6E). In addition, the expression of transcription initiation factors related to protein synthesis, including translation initiation factor 3 subunit M (*Tango7*), translation initiation factor 3 subunit B (*eIF3-S9*), and translation initiation factor 3 subunit A (*eIF3-S10*), was significantly suppressed by imidacloprid toxicity (Fig. 6F). Finally, the total levels of ATP, glycogen and protein, which are closely related to energy production and reserves, were significantly lower in developmentally delayed larvae than in controls (Fig. 6G-I).

Correlation between developmental phenotypic changes and molecular features caused by imidacloprid toxicity

To better understand the mechanism of imidacloprid-induced larval developmental retardation, we performed a comprehensive correlation analysis between epigenetic growth

traits and molecular characteristics (Fig. 7). The results showed that in larvae with imidacloprid-induced developmental retardation, body weight, width, growth index, and developmental rate were positively correlated with energy metabolism and reserves, nutrient catabolism, protein synthesis, and developmental regulation. Conversely, negative correlations were observed between these growth traits and parameters associated with detoxification and antioxidant defense processes that require additional energy expenditure. Moreover, oxidative stress and damage, detoxification, and antioxidant defense all exhibited significant positive correlations (Fig. 7).

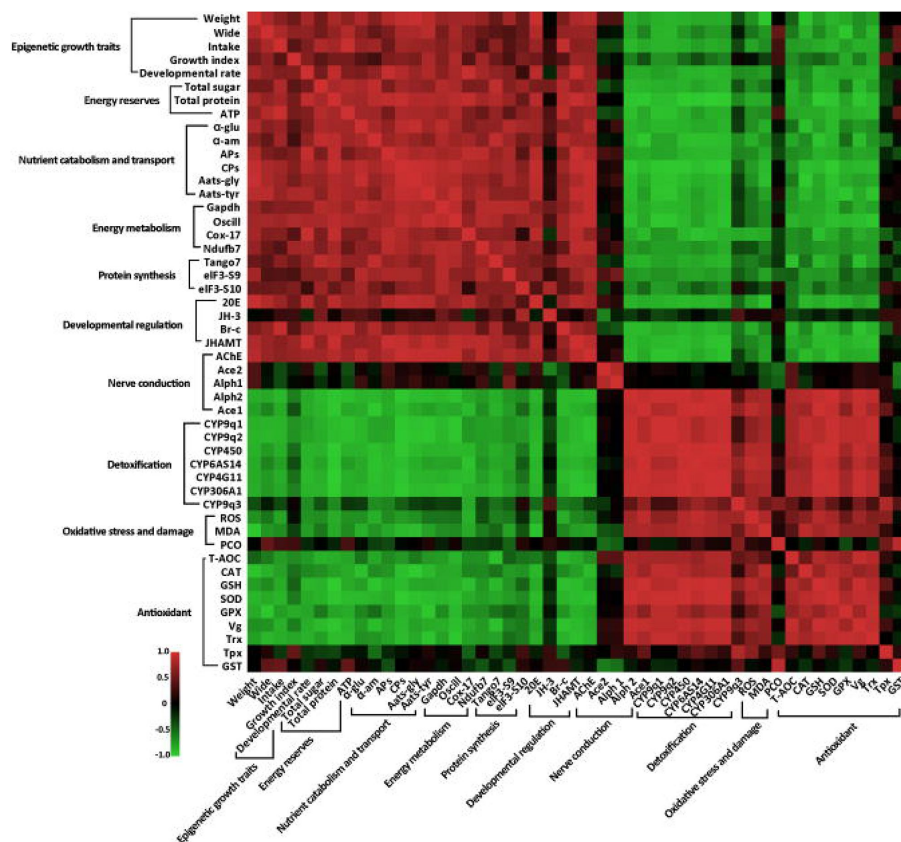


Figure 7.

Correlation between phenotypes and molecular characteristics in imidacloprid-exposed larvae with developmental retardation. Data were normalized using GraphPad software, followed by a Pearson correlation coefficient calculation using SPSS, and a heatmap was generated.

Discussion

Toxic effects and the molecular basis of environmental concentrations of imidacloprid on bee larvae

Previous research has shown that bee larvae are more susceptible to the adverse effects of pesticides than adult bees (Tomé et al., 2020). In this study, larvae were exposed to environmental concentrations of imidacloprid, resulting in sublethal effects such as decreased body weight, width, developmental rate, and growth index (Fig. 1). Our results support the notion that pesticides can affect larval development (Dai et al., 2018; Zhu et al., 2014). However, current research has primarily focused on characterizing the effects of imidacloprid on larval phenotype and behavior, and the molecular mechanisms underlying its ecotoxicological effects remain poorly understood.

To obtain molecular evidence of imidacloprid toxicity to bee larvae, we first examined the expression of acetylcholine receptors and AChE activity. The results showed that imidacloprid exposure increased the expression of the *Alph2* gene and inhibited AChE

activity (Fig. 2). Normally, acetylcholine stimulates nerve impulses by binding to the Alpha receptor, and AChE degrades acetylcholine to terminate nerve conduction. However, imidacloprid competitively binds to AChE receptors, inhibits AChE activity, and causes continuous excitation of nerve conduction, ultimately leading to paralysis and death (Katić et al., 2021; Shan et al., 2020). Therefore, our results strongly suggest that imidacloprid is neurotoxic to bee larvae.

The growth and development of juvenile animals depends on the digestion, absorption and utilization of food nutrients. Therefore, we next investigated whether imidacloprid affects the digestion and utilization of dietary by larvae. The results indicate that exposure to imidacloprid suppressed the hydrolysis genes *CPs* and *APs* and the amino acid transport pathway genes *Aats-Gly* and *Aats-Tyr* in larvae (Fig. 6B and 6C). Since these genes are responsible for cleaving dietary proteins into absorbable peptides and amino acids and transporting them for metabolism (2019), inhibiting the expression of these genes would severely affect larval digestion, resulting in protein deficiency and impaired metabolic activity. In addition, imidacloprid also affected the expression of *α-Am* and *α-Glu* genes (Fig. 6A), which are involved in carbohydrate digestion, indicating that imidacloprid also affects the digestion of dietary carbohydrates. Indeed, histological observations also confirmed that a large amount of undigested food remained in the intestine of larvae exposed to imidacloprid (Fig. 5D). In conclusion, imidacloprid inhibits the digestion and utilization of dietary proteins and carbohydrates by larvae, which is extremely detrimental to their growth and development.

In addition to nutrition, energy metabolism is critical for the maintenance of all life activities. In this study, imidacloprid exposure inhibited the expression of genes related to mitochondrial oxidative phosphorylation (*COX17*, *NDUFB7*) and its alternative glycolytic pathway (*Gapdh*, *Oscillin*). Mitochondrial oxidative phosphorylation is the central pathway for ATP generation during energy metabolism (Tian et al., 2020a), while glycolysis is an alternative mechanism for ATP synthesis (Jiang et al., 2020). Since both biochemical processes are essential for energy production, their inhibition resulted in reduced ATP levels in bee larvae (Fig. 6G). These results strongly suggest that imidacloprid induces mitochondrial dysfunction leading to insufficient energy production in larvae. Since larval energy production is insufficient, this is likely to lead to a depletion of energy reserves. Indeed, further analysis revealed a significant decrease in glycogen and protein energy reserves in larvae exposed to imidacloprid (Fig. 6H and 6I). These findings are consistent with previous reports that exposure to pesticides such as endosulfan, chlorpyrifos, insecticides, and fipronil can lead to a reduction in energy reserves in organisms (Bouayad et al., 2012a; Dutra et al., 2009; Radwan et al., 2008; Rambabu and Rao, 1994; Ribeiro et al., 2001). In conclusion, our results suggest that imidacloprid induces mitochondrial dysfunction and affects energy production and reserves in bee larvae.

Insects rely primarily on cytochrome P450 detoxification to resist toxic compounds (Li et al., 2007), and the level of CYP450 expression indicates the severity of pesticide toxicity. While adult worker bees have shown a positive response to imidacloprid (Mao et al., 2009), it is uncertain whether bee larvae can respond to pesticide toxicity. In this study, we found that seven cytochrome P450 family transcripts, including CYPq1, CYPq2, CYP450, CYP6AS14, CYP4G11, and CYP306A1, were significantly upregulated in bee larvae, with CYP9q1 being 94-fold higher than the control group (Fig. 4). These results are consistent with previous findings that the CYP450 family genes CYP6AS3, CYP6AS4, and CYP9S1 were significantly upregulated in adult worker bees exposed to quercetin, coumaphos, and fluvalinate (Mao et al., 2011) and that CYP9Q1, CYP9Q2, and CYP9Q3 help bees detoxify fluvalinate and coumaphos (Gregorc et al., 2018). This suggests that the P450 detoxification system is already established in bee larvae and they respond positively by activating the detoxification function of CYP450 upon exposure to imidacloprid.

Oxidative stress is the excessive production of ROS in aerobic organisms (Felton, 1995; Ozcan and Ogun, 2015). Pesticides, including coumaphos (Gregorc et al., 2018), fipronil (Paris et al., 2017), paraquat (Li-Byarlay et al., 2016), organophosphorus, pyrethroid, organochlorinated (Chakrabarti et al., 2015), and chlorpyrifos (Shafiq-ur-Rehman et al., 2012), have been shown to cause oxidative stress in bees. In this study, exposure to imidacloprid resulted in increased levels of ROS and MDA in larvae (Fig. 5A), indicating severe oxidative stress and lipid damage. Organisms have evolved antioxidant defense mechanisms to scavenge excessive ROS and resist oxidative stress, including increasing levels of antioxidants such as CAT, SOD, GSH, TrxR, and GSH (Dong et al., 2013; Tian et al., 2020b). In this regard, we have recently confirmed this in worker bees (Li et al., 2022), which is also well reflected in the present study. We observed a significant increase in the activities of antioxidant enzymes SOD and CAT, gene expression of Trx and Gpx, and GSH levels in larvae after exposure to imidacloprid (Fig. 5B and 5C). These results show that the antioxidant system in developing larvae is established and used to resist oxidative stress caused by imidacloprid, similar to that in adult worker bees. Notably, compared to our recent study in adult bees (Li et al., 2022), the larvae in this study exhibited more robust antioxidant activity when exposed to higher concentrations of imidacloprid toxicity than worker bees. However, the underlying mechanism requires further investigation. The underlying mechanisms involved require further study.

Notably, in addition to routine antioxidants, imidacloprid induces a 7-fold increase in Vg expression (Fig. 3A). Vg is known to provide nutrition to developing embryos (Jorgensen et al., 2009) and to help animals cope with pollution-induced stress (Matozzo et al., 2008). In bees, Vg plays an important antioxidant role by protecting other hemolymph molecules from ROS and increasing cellular oxidative stress tolerance (Corona et al., 2007). Previous studies have shown a correlation between high Vg expression in adult worker bees and resistance to oxidative stress resulting from exposure to the pesticide paraquat (Seehuus et al., 2006). In this study, our results support this observation and suggest that larvae may also resist oxidative stress caused by imidacloprid through increased Vg production.

Mechanisms of imidacloprid-induced developmental retardation in bee larvae

Impaired juvenile development can have a significant impact on the growth and sustainability of entire populations. Deviations in juvenile development due to environmental stress have recently received considerable attention. *Drosophila melanogaster* exposed to heat stress exhibit developmental delays due to the failure of early embryos to fully inhibit the synthesis of non-heat shock proteins (Bergh and Arking, 1984). Similarly, exposure to nutrient and heavy metal stress can cause delayed or arrested development in *Caenorhabditis elegans* due to altered gene expression and disruption of developmental signaling pathways (Carranza-García and Navarro, 2020; Rashid et al., 2021). In the present study, we also observed typical developmental delays in honeybee larvae exposed to imidacloprid, but the underlying mechanisms remain unclear.

Insect development is regulated by the hormones JH and 20E, which control morphological remodeling events during molting. 20E promotes pupation (Yuan et al., 2020), while JH maintains juvenile characteristics and prevents metamorphosis by antagonizing the effects of 20E (Luo et al., 2021). To elucidate the mechanism underlying imidacloprid-induced larval developmental delay, we first focused on the hormonal regulation of molting development and found that imidacloprid inhibits the expression of JHAMT (Fig. 3A), a positive regulator of JH synthesis (Liu et al., 2018), but did not affect JH titer levels (Fig. 3B). This suggests that the antagonistic effect of JH still restricts the developmental process, thereby maintaining larval morphology. In parallel, imidacloprid toxicity also resulted in reduced ecdysteroid 20E

titer and *Br-c* expression (Fig. 3). The role of 20E is to induce larval molting and pupation by triggering downstream response factors, such as *Br-c* (Deng et al., 2012). Therefore, our results suggest that larval developmental delay may be caused by imidacloprid reducing 20E titer and inhibiting *Br-c* expression, thereby blocking molting. Notably, we also observed that a decrease in 20E levels was accompanied by a decrease in AChE activity (Fig. 2A), which is known to cause delayed pupation (He et al., 2012) and developmental arrest (Desneux et al., 2007) in insect larvae. Therefore, our results suggest that imidacloprid neurotoxicity may cause developmental delay in bee larvae by inhibiting the molting hormone 20E. This conclusion is supported by a correlation analysis showing that decreases in 20E titer, *Br-c* expression and AChE activity were positively correlated with decreased developmental rates and growth index, weight and width growth (Fig. 7). In conclusion, imidacloprid neurotoxicity inhibits the 20E and *Br-c* genes, thereby blocking molting, which may be an important cause of delayed larval development in honey bees.

Animals employ selective foraging as a survival strategy to avoid ingesting toxic foods that may cause adverse physiological effects (Berenbaum and Johnson, 2015). Although bees cannot actively reject toxins innately, postingestive malaise allows them to learn to avoid ingesting toxic-containing foods with adverse physiological effects (Hurst et al., 2014). Given this, we speculate that it is likely that the honey bee larvae in this study reduced their intake of toxic imidacloprid-containing foods, making it difficult to meet their nutritional and energy needs, which may have resulted in delayed development. However, monitoring of daily food intake did not show a significant difference between larvae exposed to imidacloprid and the control group (Fig. 1F and 1G), indicating that larval food intake was not reduced and larvae did not initiate avoidance behavior toward toxic food. Therefore, food intake was not a critical factor in the delayed larval development caused by imidacloprid.

Nutrition and energy are critical for the growth and development of juvenile animals. Exposure to heavy metals and neonicotinoid pesticides has been shown to affect digestion, energy reserves, production, and utilization in animals (Jiang et al., 2020; Li et al., 2021). Given this, since the lack of the necessary correlation between delayed larval development and food intake in this study prompts us to consider potential problems in larval food and energy utilization that led to delayed development. Consequently, our focus shifted to the investigation of larval nutrition and energy metabolism. Interestingly, we found severe pathological changes in the gut tissues of developmentally delayed larvae exposed to imidacloprid (Fig. 5D). Imidacloprid caused a significant reduction in the tightness of the cell arrangement in the basal layer of the gut, and a drastic decrease in the number of cells. The peritrophic membrane was incompletely formed, and undigested food residues were present in the gut. qRT-PCR analysis revealed a significant decrease in larval digestion and catabolism of dietary proteins and carbohydrates, amino acid transport, oxidative phosphorylation, and glycolysis (Fig. 6D-6F). Biochemical assays showed that imidacloprid stress also decreased ATP levels and total glycogen and protein content, resulting in a lack of energy reserves and increased energy expenditure (Fig. 6G-6I). Correlation analysis indicated a positive relationship between impaired catabolism and dietary nutrient utilization caused by imidacloprid and larval developmental delay (Fig. 7). These evidence strongly suggest that imidacloprid exposure affects the digestion and utilization of nutrients and energy in larvae, resulting in insufficient nutrients and energy for growth and development, which may be another important factor contributing to imidacloprid-induced larval developmental delay.

The Dynamic Energy Budget (DEB) theory suggests that environmental stress can significantly affect the energy balance of organisms, leading to increased energy expenditure (Kooijman, 2009). To cope with stress, animals may deplete their energy reserves, such as glycogen and protein (Bouayad et al., 2012b; Matsukura et al., 2008). In this

study, we found that larvae with delayed development caused by imidacloprid had lower energy reserves (ATP, total protein, and total glycogen), and these energy reserves were negatively correlated with larval antioxidant and detoxification (Fig. 7), suggesting that the reduction in energy reserves caused by imidacloprid is related to the energy-consuming activities of CYP450 detoxification and antioxidant defense. To meet the high energy cost demands of environmental stress, animals often have to allocate most of their food intake to defense (Guedes et al., 2006), and this redistribution of energy resources will inevitably lead to an insufficient energy supply for growth and development (Beyers et al., 1999). Therefore, the P450 detoxification and antioxidant defense used by the larvae in this study to resist imidacloprid toxicity will inevitably force them to consume more food resources that should have been used for growth and development, resulting in a severe lack of energy needed for development. This may also be another important reason why imidacloprid causes delayed development in bee larvae.

Conclusion

Imidacloprid toxicity caused growth and developmental delay in *A. mellifera* larvae. The toxic mechanism may include imidacloprid disrupting the regulatory balance of molt development, resulting in restricted development. In addition, the gut damage caused by imidacloprid restricts the metabolism and utilization of food nutrients and energy in the larvae. Third, the additional energy consumed by larval P450 detoxification and antioxidant defenses further reduces the supply of nutrients and energy for growth and development (Fig. 8). This study is the first multi-level investigation of the toxic effects of imidacloprid on *A. mellifera* larvae. These findings have broader implications for understanding and assessing the threat of harmful pesticides to animal larvae.

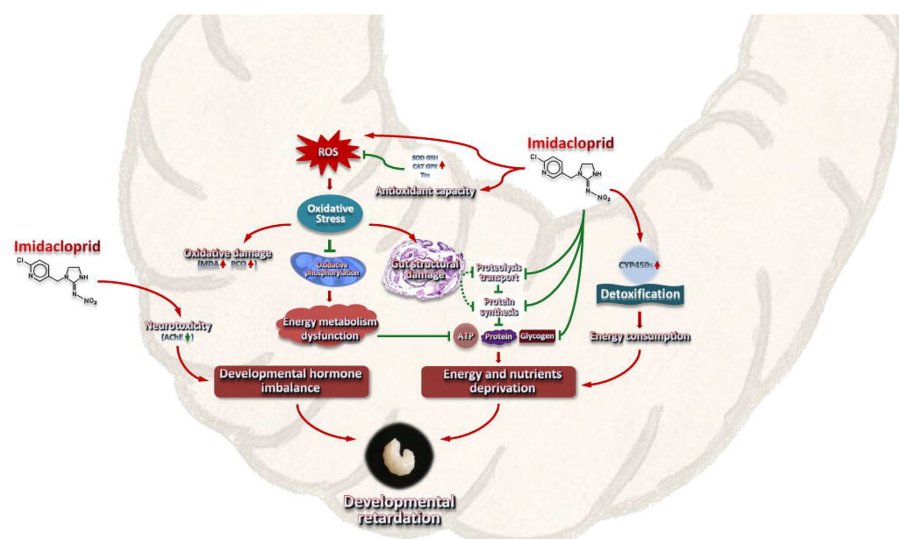


Figure 8.

A model interpreting the ecotoxicological effects of imidacloprid on *Apis mellifera* larvae and the mechanisms of imidacloprid-induced developmental retardation. Imidacloprid toxicity caused growth and developmental delay in *A. mellifera* larvae. It disrupts the regulatory balance of molt development, resulting in restricted development. In addition, the damage caused by imidacloprid restricts the metabolism and utilization of food nutrients and energy in larvae. The additional energy consumed by larval P450 detoxification

and antioxidant defenses further reduces the supply of nutrients and energy supply for growth and development. The arrows indicate induction or promotion, the straight lines indicate inhibition, and the green dotted lines indicate speculative inhibition.

Materials and Methods

Larval rearing

The Animal Bioethics Committee at Chongqing Normal University, China approved all experimental protocols. The chosen colonies were healthy and not exposed to pathogens or pesticides. Two-day-old larvae were individually transferred to sterile 24-well cell culture plates filled with a standard diet consisting of royal jelly, glucose, fructose, water, and yeast extract (Aupinel et al., 2005). The larvae were cultured in dark conditions at 35°C and 96% RH, with daily feeding volumes adjusted according to their age (20 µL for three-day-old larvae, 30 µL for four-day-old larvae, 40 µL for five-day-old larvae, and 50 µL for six-day-old larvae). Any remaining food was recorded and removed.

Experimental design

This study employed four different concentrations of imidacloprid, which were residual concentrations found in honey (0.7 ppb) (Chauzat et al., 2009), bees (1.2 ppb) (Chauzat et al., 2011), pollen (3.1 ppb) (Mullin et al., 2010), and beeswax (377 ppb) (Kapoor et al., 2014). The experiment was conducted with three treatment groups: control group (CK), solvent control group (CKac), and imidacloprid-treated group (IMI), each with three replicates. Each replicate contained 32 bees that were distributed in standard 24-well cell culture plates (1 bee/well). The CK group consisted of larval bees fed acetone-royal jelly glucose-fructose-yeast extract-distilled water (0.00377%-50%-6%-6%-1%-36.99623%, v/w/w/w/w/v). The IMI group consisted of larvae-fed control solution containing imidacloprid (99% active ingredient, Hubei Norna Technology Co. Ltd, Hubei, China) at 0.7 ppb, 1.2 ppb, 3.1 ppb, and 377 ppb. The CKac group served as a solvent control and contained only acetone at a concentration of 0.00377%. A total of 312 larvae were used for survival statistics (96 bees), body weight and width (96 bees), and gene expression, enzyme activity, histochemical content, and histopathologic analysis. Food was administered daily according to age (20 µL for three-day-old larvae, 30 µL for four-day-old larvae, 40 µL for five-day-old larvae, and 50 µL for six-day-old larvae). Dead individuals and food remain were recorded daily, and incubation conditions were 35°C and 96% RH. Surviving larvae were collected 72 h after consumption of the test solution and stored at -80°C for further analysis, including evaluations of gene expression, enzyme activity, and chemical content evaluations. Furthermore, HE-stained sections were used to evaluate any imidacloprid-induced tissue damage and apoptosis.

Developmental phenotypic traits

Larval individuals were randomly selected daily from the CK and IMI groups to measure body weight and width. Food residues, larval mortality, developmental progression, and the amount of time spent in the larval developmental stage (from the first imidacloprid exposure to before pupa) were recorded daily to calculate food consumption, survival rates, developmental rates, and development times. The growth index (GI) was calculated for larvae exposed to imidacloprid from the three-day-old to the six-day-old using the equations described by Zhang (Zhang et al., 1993). The developmental rate was measured as the number of larvae successfully reaching the pupal stage as a percentage of the total initial sample size.

RNA extraction, cDNA synthesis, and real-time quantitative PCR analysis

Total RNA was extracted from larvae using TRIzol reagent (Thermo Fisher Scientific) and RNAex Pro Reagent (Accurate Biology, China) as instructed by the manufacturer. The RNA quality was evaluated by gel electrophoresis and absorbance determination using a NanoDrop 2000 spectrophotometer (Thermo Scientific, New York, USA). Genomic DNA was removed, and cDNA was synthesized using a reverse transcription kit (Evo M-MLV RT Mix Kit with gDNA Clean for qPCR, Accurate Biology, China) according to the manufacturer's instructions. Briefly, 1 µg of RNA reacted with the 5X gDNA Clean Reaction Mix provided in the kit at 42°C for 5 min to remove residual genomic DNA to ensure the accuracy of the quantitative results. After the reaction, 4 µL of 5X Evo M-MLV RT Reaction Mix and 4 µL of RNase-free water were added to the reaction tube for 20 µL. The mixture was incubated at 37°C for 15 min and 85°C for 5 sec to generate cDNA. Then, quantitative RT-PCR was performed with a Bio-Rad CFX96™ Real-time PCR Detection System (Bio-Rad, Hercules, CA, USA) according to standard protocols and cycling conditions. RP49 (ribosomal protein 49, Accession no. AF441189) (Lourenco et al., 2008) was used as the endogenous control. The primers for target genes and RP49 are listed in Table S1. qPCR system (20 µL): EvaGreen Express 2X qPCR MasterMix (10 µL), each forward and reverse primer (0.4 µL, 0.2 µmol/L), cDNA (1 µL, 100 ng/µL), ddH₂O (8.2 µL). The thermal cycling program was as follows: 95°C for 30 sec, followed by 40 cycles of 95°C for 5 s and 60°C for 30 s. The qPCR products were sequenced by Shanghai Biological Engineering Technology Services Co., Ltd. to confirm whether the amplified fragments were the target sequences. Relative mRNA expression levels of target genes were calculated using the 2^{-ΔΔC_t} method.

Histochemical measurement

Juvenile hormone 3 (JH-III), 20-hydroxyecdysone (20E) titers, and reactive oxygen species (ROS) levels in larvae were determined using ELISA enzyme-linked immunosorbent assay kits (Jining, Shanghai, China) according to the manufacturer's instructions. Briefly, the larva was homogenized on ice in 5 volumes (M: V) of phosphate-buffered saline (PBS) buffer, and the supernatant was collected after centrifugation at 12,000 x g for 20 min. Subsequently, 50 µL of reference standard solution or 10 µL of sample supernatant plus 40 µL of dilution were added to the microtiter wells and incubated at 37°C for 60 min with 100 µL of HRP-labeled detection antibody. Each well was washed five times with 400 µL of PBS buffer containing 0.05% Tween 20, pat dried, and left to stand at room temperature for 5 min. TMB chromogen solution A (50 µL) and solution B (50 µL) were added to each well, gently mixed, and incubated at 37°C for 15 min in the dark. Then, 50 µL of H₂SO₄ (4 mol/L) was added to each well to terminate the reaction. Finally, the OD value was measured at 450 nm with a NanoDrop 2000 spectrophotometer (Thermo Scientific, New York, USA), and the content of the detected substance was calculated from a calibration curve constructed by reference standards.

To prepare samples for analysis of protein carbonylation (PCO), acetylcholinesterase (AChE), catalase (CAT), malondialdehyde (MDA), total antioxidant capacity (T-AOC), superoxide dismutase (SOD), and glutathione (GSH), larvae was weighed and homogenized on ice in 5 volumes (M: V) of PBS buffer, then centrifuged at 12,000 x g for 15 min to collect the supernatant. Subsequently, PCO content was measured using the 2,4-dinitrophenylhydrazine (DNPH) method with slight modifications. The above-prepared supernatant was mixed with streptomycin sulfate at a 9:1 volume (V: M), kept at room temperature for 10 min, and then centrifuged at 12,000 x g for 15 min. Then, 80 µL of supernatant was mixed with 160 µL of 10LrM DNPH in 2LrM HCl and reacted at 37°C for 30 min. Sample controls

were also prepared by adding an equal volume (500 μ L) of 2 LrM HCl. Then, 200 μ L of 20% TCA was added, mixed thoroughly, and centrifuged at 4°C for 10 min at 12,000 $\times$ g, and the precipitate was collected. Add 400 μ L of ethanol: ethyl acetate mixture (1:1), vortex and mix, centrifuge at 4°C 12,000 $\times$ g for 10 min, and collect the precipitate to remove unbound DNPH. This DNPH removal step was repeated once. The pellet was mixed with 400 μ L of 6 LrM guanidine hydrochloride, kept at room temperature for 15 min, and then centrifuged at 4°C 12,000 $\times$ g for 10 min to collect the supernatant. The carbonyl and protein contents were determined by reading the absorbance at $\lambda_{\text{Lr}} = \text{Lr}370\text{Lrnm}$ and 280 nm with a SpectraMax 190 spectrophotometer (Molecular Devices, California, USA) for each sample against an appropriate blank. Finally, the carbonyl content per unit of protein was calculated based on the protein content.

AChE activity was measured by the Ellman method as described by Orhan et al. (Orhan et al., 2007) with slight modifications. Briefly, 20 μ L of the prepared sample supernatant was mixed with 160 μ L of 0.1 mM PBS buffer (pH 8.0) and 10 μ L of an AChE (0.28 U mL) and incubated for 15 min at room temperature. Then, 10 μ L of ATChI (acetylthiocholine iodide, 4 mg mL) was added together with 20 μ L of DTNB (5,5-dithiobis-2-nitrobenzoic acid, 1.2 mg mL) and immediately mixed, and the absorbance was read at 412 nm. The results were expressed as activity percentages relative to the sample solvent negative control.

The CAT activity was determined using a kit (Beyotime, Shanghai, China) according to the manufacturer's instructions with slight modifications. Briefly, 10 μ L of the prepared sample supernatant was mixed with 30 μ L of Tris-HCl buffer (pH 7.0, 50 mmol L), 10 μ L of 250 mM H_2O_2 was added, and the mixture was immediately mixed and kept at 25°C for 5 min. Then, 450 μ L of 1.8 mol L⁻¹ H_2SO_4 was added, and the mixture was vortexed to terminate the reaction. Subsequently, 10 μ L of this termination reaction solution was mixed with 40 μ L of Tris-HCl buffer (pH 7.0, 50 mmol L). Then, 10 μ L of this mixture was added to a 96-well plate mixed with 200 μ L of color development working solution containing peroxidase and incubated for 15 min at 25°C. Finally, the absorbance was measured at 520 nm, and the CAT activity was calculated using a calibration curve of 5 mM H_2O_2 .

T-AOC was measured using the total antioxidant capacity assay kit with the ABTS (2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid) method based on a previous report (Chen et al., 2019). Add 10 μ L of the prepared sample supernatant to 200 μ L of ABTS working solution. After 6 min of reaction, the absorbance was measured at 734 nm. Finally, the antioxidant activity was displayed by Trolox equivalents antioxidant capacity as mmol Trolox equivalents/g extract. The ABTS working solution was prepared as follows: 400 μ L of potassium persulfate was added to 400 μ L of ABTS to generate an ABTS stock solution, which was then incubated for 16 h at room temperature in the dark. Before use, the stock solution was diluted with phosphate buffer, and the absorbance at 734 nm was adjusted to 0.70 ± 0.05 .

Additionally, MDA, SOD, and GSH levels were detected using ELISA enzyme-linked immunosorbent assay kits (Beyotime, Shanghai, China) according to our recent study (Li et al., 2022).

Histopathological analysis

Microsectioning and hematoxylin-eosin (HE) staining was performed according to standard histological protocols (Khan et al., 2010) with some modifications. Briefly, entire fresh larvae were fixed in 10% neutral-buffered formalin solution for 12 h at 4°C after 72 h of imidacloprid exposure. After isopropyl alcohol dehydration, samples were embedded in paraffin and sectioned. Three-micron-thick sections were stained with a Hematoxylin and

Eosin Staining Kit (Beyotime, Shanghai, China), observed with an Olympus BX51 microscope, and photographed with an Olympus digital camera.

Statistical analysis

Data were obtained from three independent experiments performed in triplicate and are presented as the mean $\pm$ SEM. SPSS statistical package was used for statistical analysis using Student's t-test and one-way ANOVA followed by Tukey's multiple comparison test (* $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$ were considered statistically significant). GraphPad software was used for data normalization, Pearson correlation coefficient calculation was analyzed using SPSS, and a heat map was generated.

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Additional information

Author contributions

Zhi Li: Supervision, Designed research, Conceptualization, Methodology, Funding acquisition, Writing-original draft, Writing-review & editing. *Yuedi wang*: Investigation, Methodology, Validation, Formal analysis, Project administration. *Lanchun Chen*: Investigation, Data curation. *Qiqian Qing*: Investigation, Data curation. *Xiaoqun Dang*: Data curation. *Zhengang Ma*: Data curation. *Zeyang Zhou*: Conceived the research, Research guidance. All the authors read, corrected, and approved the manuscript.

Ethics

Animal experimentation: This study was performed in strict accordance with the guidelines of the Animal Bioethics Committee at Chongqing Normal University, China approved all experimental protocols.

Data Availability

All the data needed to evaluate the conclusions in this paper are present in the paper and/or Supplementary Materials.

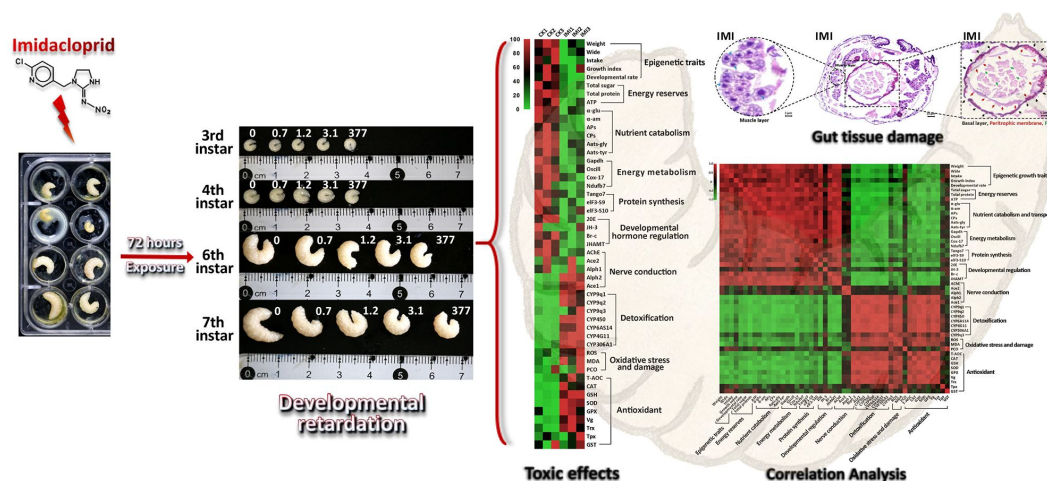
Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional files

Supplementary files: Table S1. Primers used in real-time quantitative PCR.

Imidacloprid toxicity and the underlying mechanism causing larval development delay



References

- (2019) **A Review on Microbial Alkaline Protease: An Essential Tool for Various Industrial Approaches** *Industrial Biotechnology* **15**:69–78
- Anatra-Cordone M D. P (2005) **Imidacloprid. Human health assessment and ecological risk assessment Final report.** . Syracuse Environmental Research Associates Inc, New York *SERA TR* :5–43
- Anderson N. L. , Harmon-Threath A. N (2021) **Chronic contact with imidacloprid during development may decrease female solitary bee foraging ability and increase male competitive ability for mates** *Chemosphere* **283**
- Aupinel P. , et al. (2005) **Improvement of artificial feeding in a standard in vitro method for rearing Apis mellifera larvae** *Bulletin of Insectology* **58**:107–111
- Belzunces L. P. , et al. (2012) **Neural effects of insecticides in the honey bee.** Springer-Verlag
- Berenbaum M. R. , Johnson R. M. (2015) **Xenobiotic detoxification pathways in honey bees** *Curr Opin Insect Sci.* **10**:51–58
- Bergh S. , Arking R (1984) **Development profile of the heat shock response in early embryos of Drosophila** *J Exp Zool* **231**:379–91

Beyers D. W. , et al. (1999) **Estimating physiological cost of chemical exposure: integrating energetics and stress to quantify toxic effects in fish** *Canadian Journal of Fisheries and Aquatic Sciences* **56**:814–822

Bhme F. , et al. (2018) **From field to food—will pesticide-contaminated pollen diet lead to a contamination of royal jelly?** Springer Paris

Bouayad N. , et al. (2012) **Dietary effects of harmine, a beta-carboline alkaloid, on development, energy reserves and alpha-amylase activity of *Plodia interpunctella* Hubner (Lepidoptera: Pyralidae)** *Saudi J Biol Sci* **19**:73–80

Bouayad N. , et al. (2012) **Dietary effects of harmine, a β -carboline alkaloid, on development, energy reserves and α -amylase activity of *Plodia interpunctella* Hübner (Lepidoptera: Pyralidae)** *Saudi J Biol Sci* **19**:73–80

Cardoso P. , et al. (2020) **Scientists' warning to humanity on insect extinctions** *Biological Conservation* **242**

Carneiro L. S. , et al. (2022) **Acute oral exposure to imidacloprid induces apoptosis and autophagy in the midgut of honey bee *Apis mellifera* workers** *Sci Total Environ* **815**

Carranza-García E. , Navarro R. E (2020) **Insights Into the Hypometabolic Stage Caused by Prolonged Starvation in L4-Adult *Caenorhabditis elegans* Hermaphrodites** *Front Cell Dev Biol* **8**

Chaimanee V. , et al. (2016) **Sperm viability and gene expression in honey bee queens (*Apis mellifera*) following exposure to the neonicotinoid insecticide imidacloprid and the organophosphate acaricide coumaphos** *J Insect Physiol* **89**:1–8

Chakrabarti P. , et al. (2015) **Pesticide-induced oxidative stress in laboratory and field populations of native honey bees along intensive agricultural landscapes in two Eastern Indian states** *Apidologie*

Chauzat M. P. , et al. (2009) **Influence of pesticide residues on honey bee (Hymenoptera: Apidae) colony health in France** *Environ Entomol* **38**:514–23

Chauzat M. P. , et al. (2011) **An assessment of honeybee colony matrices, *Apis mellifera* (Hymenoptera: Apidae) to monitor pesticide presence in continental France** *Environ Toxicol Chem* **30**:103–11

Chen J. , et al. (2019) **Cytoprotective Effect of *Ligustrum robustum* Polyphenol Extract against Hydrogen Peroxide-Induced Oxidative Stress via Nrf2 Signaling Pathway in Caco-2 Cells** *Evid Based Complement Alternat Med* **2019**

Corona M. , et al. (2007) **Vitellogenin, juvenile hormone, insulin signaling, and queen honey bee longevity** *Proc Natl Acad Sci U S A* **104**:7128–33

- Dai P. , et al. (2018) **The Herbicide Glyphosate Negatively Affects Midgut Bacterial Communities and Survival of Honey Bee during Larvae Reared in Vitro** *J Agric Food Chem* **66**:7786–7793
- Deng H. , et al. (2012) **Homeodomain POU and Abd-A proteins regulate the transcription of pupal genes during metamorphosis of the silkworm, Bombyx mori** *Proc Natl Acad Sci U S A* **109**:12598–603
- Desneux N. , et al. (2007) **The sublethal effects of pesticides on beneficial arthropods** *Annu Rev Entomol* **52**:81–106
- Dively G. P. , Kamel A (2012) **Insecticide residues in pollen and nectar of a cucurbit crop and their potential exposure to pollinators** *J Agric Food Chem* **60**:4449–56
- Dong M. , et al. (2013) **Toxic effects of 1-decyl-3-methylimidazolium bromide ionic liquid on the antioxidant enzyme system and DNA in zebrafish (Danio rerio) livers** *Chemosphere* **91**:1107–12
- Dutra B. K. , et al. (2009) **Carbofuran-induced alterations in the energy metabolism and reproductive behaviors of Hyalella castroi (Crustacea Amphipoda).** *Comp Biochem Physiol C Toxicol Pharmacol* **149**:640–6
- Gill R. J. , et al. (2012) **Combined pesticide exposure severely affects individual- and colony-level traits in bees** *Nature* **491**:105–8
- Gregorc A. , et al. (2018) **Effects of coumaphos and imidacloprid on honey bee (Hymenoptera: Apidae) lifespan and antioxidant gene regulations in laboratory experiments** *Sci Rep* **8**
- Guedes R. N. C. , et al. (2006) **Cost and mitigation of insecticide resistance in the maize weevil, Sitophilus zeamais** *Physiological Entomology* **31**:30–38
- He G. , et al. (2012) **RNA interference of two acetylcholinesterase genes in Plutella xylostella reveals their different functions** *Archives of Insect Biochemistry and Physiology* **79**:75–86
- Hurst V. , et al. (2014) **Toxins induce ‘malaise’ behaviour in the honeybee (Apis mellifera)** *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* **200**:881–90
- Islam M. A. , et al. (2019) **Acute Toxicity of Imidacloprid on the Developmental Stages of Common Carp Cyprinus carpio** *Toxicology and Environmental Health Sciences* **11**:244–251
- Jeschke P. , et al. (2011) **Overview of the status and global strategy for neonicotinoids** *J Agric Food Chem* **59**:2897–908
- Jia Q. , Li S (2023) **Mmp-induced fat body cell dissociation promotes pupal development and moderately averts pupal diapause by activating lipid metabolism** *Proc Natl Acad Sci U S A* **120**

- Jiang D. , et al. (2020) **Cd exposure-induced growth retardation involves in energy metabolism disorder of midgut tissues in the gypsy moth larvae** *Environ Pollut* **266**
- Jorgensen P. , et al. (2009) **The mechanism and pattern of yolk consumption provide insight into embryonic nutrition in *Xenopus*** *Development* **136**:1539–48
- Kapoor U. , et al. (2014) **Disposition and acute toxicity of imidacloprid in female rats after single exposure** *Food Chem Toxicol* **68**:190–5
- Karahan A. , et al. (2015) **Sublethal imidacloprid effects on honey bee flower choices when foraging** *Ecotoxicology* **24**:2017–25
- Katić A. , et al. (2021) **Effects of low-level imidacloprid oral exposure on cholinesterase activity, oxidative stress responses, and primary DNA damage in the blood and brain of male Wistar rats** *Chemico-Biological Interactions* **338**
- Khan D. A. , et al. (2010) **Monitoring health implications of pesticide exposure in factory workers in Pakistan** *Environ Monit Assess* **168**:231–40
- Kooijman B (2009) **Dynamic Energy Budget Theory for Metabolic Organisation**
- Li-Byarlay H. , et al. (2016) **Honey bee (*Apis mellifera*) drones survive oxidative stress due to increased tolerance instead of avoidance or repair of oxidative damage** *Exp Gerontol* **83**:15–21
- Li X. , et al. (2007) **Molecular mechanisms of metabolic resistance to synthetic and natural xenobiotics** *Annu Rev Entomol* **52**:231–53
- Li X. , et al. (2021) **New insights into crosstalk between apoptosis and necroptosis co-induced by chlorothalonil and imidacloprid in *Ctenopharyngodon idellus* kidney cells** *Sci Total Environ* **780**
- Li Z. , et al. (2022) **Melatonin enhances the antioxidant capacity to rescue the honey bee *Apis mellifera* from the ecotoxicological effects caused by environmental imidacloprid** *Ecotoxicol Environ Saf* **239**
- Liu D. , et al. (2018) **Toxicity and sublethal effects of fluralaner on *Spodoptera litura* Fabricius (Lepidoptera: Noctuidae)** *Pestic Biochem Physiol* **152**:8–16
- Lourenco A. P. , et al. (2008) **Validation of reference genes for gene expression studies in the honey bee, *Apis mellifera*, by quantitative real-time RT-PCR** *Apidologie* **39**:372–385
- Luo W. , et al. (2021) **Juvenile hormone signaling promotes ovulation and maintains egg shape by inducing expression of extracellular matrix genes** *Proceedings of the National Academy of Sciences* **118**
- Mahé C. , et al. (2021) **The countryside or the city: Which environment is better for the honeybee?** *Environ Res* **195**

- Mao W. , et al. (2009) **Quercetin-metabolizing CYP6AS enzymes of the pollinator *Apis mellifera* (Hymenoptera: Apidae)** *Comp Biochem Physiol B Biochem Mol Biol* **154**:427–34
- Mao W. , et al. (2011) **CYP9Q-mediated detoxification of acaricides in the honey bee (*Apis mellifera*)** *Proc Natl Acad Sci U S A* **108**:12657–62
- Matozzo V. , et al. (2008) **Vitellogenin as a biomarker of exposure to estrogenic compounds in aquatic invertebrates: a review** *Environ Int* **34**:531–45
- Matsukura K. , et al. (2008) **Changes in chemical components in the freshwater apple snail, *Pomacea canaliculata* (Gastropoda: Ampullariidae), in relation to the development of its cold hardiness** *Cryobiology* **56**:131–7
- Medrzycki P. , et al. (2003) **Effects of imidacloprid administered in sub-lethal doses on honey bee behaviour** *Laboratory tests. Bulletin of Insectology* **56**:59–62
- Mullin C. A. , et al. (2010) **High levels of miticides and agrochemicals in North American apiaries: implications for honey bee health** *PLoS One* **5**
- Orhan I. , et al. (2007) **Antioxidant and anticholinesterase evaluation of selected Turkish *Salvia* species** *Food Chemistry* **103**:1247–1254
- Paris L. , et al. (2017) **Disruption of oxidative balance in the gut of the western honeybee *Apis mellifera* exposed to the intracellular parasite *Nosema ceranae* and to the insecticide fipronil** *Microb Biotechnol* **10**:1702–1717
- Peng Y. C. , Yang E. C (2016) **Sublethal Dosage of Imidacloprid Reduces the Microglomerular Density of Honey Bee Mushroom Bodies** *Sci Rep* **6**
- Pettis J. S. , et al. (2013) **Crop pollination exposes honey bees to pesticides which alters their susceptibility to the gut pathogen *Nosema ceranae*** *PLoS One* **8**
- Radwan M. A. , et al. (2008) **Biochemical and histochemical studies on the digestive gland of *Eobania vermiculata* snails treated with carbamate pesticides** *Pesticide Biochemistry and Physiology* **90**:154–167
- Rambabu J. P. , Rao M. B (1994) **Effect of an organochlorine and three organophosphate pesticides on glucose, glycogen, lipid, and protein contents in tissues of the freshwater snail *Bellamya dissimilis* (Muller)** *Bull Environ Contam Toxicol* **53**:142–8
- Rashid S. , et al. (2021) **Developmental plasticity and the response to nutrient stress in *Caenorhabditis elegans*** *Developmental Biology*
- Ribeiro S. , et al. (2001) **Effect of endosulfan and parathion on energy reserves and physiological parameters of the terrestrial isopod *Porcellio dilatatus*** *Ecotoxicol Environ Saf* **49**:131–8

- Rocha C. M. , et al. (2022) **Environmental photochemical fate of pesticides ametryn and imidacloprid in surface water (Paranapanema River, Sao Paulo Brazil).** *Environ Sci Pollut Res Int* **29**:42290–42304
- Samojeden C. G. , et al. (2022) **Toxicity and genotoxicity of imidacloprid in the tadpoles of *Leptodactylus luctator* and *Physalaemus cuvieri* (Anura: Leptodactylidae)** *Scientific Reports* **12**
- Sánchez-Bayo F. , Wyckhuys K. A. G (2019) **Worldwide decline of the entomofauna: A review of its drivers** *Biological Conservation* **232**:8–27
- Seehuus S.-C. , et al. (2006) **Reproductive protein protects functionally sterile honey bee workers from oxidative stress** *Proceedings of the National Academy of Sciences* **103**:962–967
- Shafiq-ur-Rehman , et al. (2012) **Chlorpyrifos-induced neuro-oxidative damage in bee** *Toxicology & Environmental Health Sciences* **4**:30–36
- Shan Y. , et al. (2020) **Effect of imidacloprid on the behavior, antioxidant system, multixenobiotic resistance, and histopathology of Asian freshwater clams (*Corbicula fluminea*)** *Aquat Toxicol* **218**
- Shaw G. M. , et al. (2014) **Early pregnancy agricultural pesticide exposures and risk of gastroschisis among offspring in the San Joaquin Valley of California** *Birth Defects Res A Clin Mol Teratol* **100**:686–94
- Simon-Delso N. , et al. (2015) **Systemic insecticides (neonicotinoids and fipronil): trends, uses, mode of action and metabolites** *Environ Sci Pollut Res Int* **22**:5–34
- Tian F. , et al. (2020) **Effects of Thymoquinone on Small-Molecule Metabolites in a Rat Model of Cerebral Ischemia Reperfusion Injury Assessed using MALDI-MSI** *Metabolites* **10**
- Tian X. , et al. (2020) **Neonicotinoids caused oxidative stress and DNA damage in juvenile Chinese rare minnows (*Gobiocypris rarus*)** *Ecotoxicol Environ Saf* **197**
- Tomé H. V. V. , et al. (2020) **Frequently encountered pesticides can cause multiple disorders in developing worker honey bees** *Environmental Pollution* **256**
- Tong Z. , et al. (2018) **A survey of multiple pesticide residues in pollen and beebread collected in China** *Sci Total Environ* :640–641
- Tudi M. , et al. (2021) **Agriculture Development, Pesticide Application and Its Impact on the Environment** *Int J Environ Res Public Health* **18**
- Yang E. C. , et al. (2012) **Impaired olfactory associative behavior of honeybee workers due to contamination of imidacloprid in the larval stage** *PLoS One* **7**

Yang W. , et al. (2014) **Residential agricultural pesticide exposures and risk of neural tube defects and orofacial clefts among offspring in the San Joaquin Valley of California** *Am J Epidemiol* **179**:740–8

Yuan D. , et al. (2020) **The AMPK-PP2A axis in insect fat body is activated by 20-hydroxyecdysone to antagonize insulin/IGF signaling and restrict growth rate** *Proceedings of the National Academy of Sciences* **117**:9292–9301

Zhang M. , et al. (1993) **Quantification of insect growth and its use in screening of naturally occurring insect control agents** *J Chem Ecol* **19**:1109–18

Zheng H. Q. , Fu-Liang H. U (2009) **Honeybee: a newly emerged model organism** *Acta Entomologica Sinica* :210–215

Zhu W. , et al. (2014) **Four common pesticides, their mixtures and a formulation solvent in the hive environment have high oral toxicity to honey bee larvae** *PLoS One* **9**

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Reviewer #1 (Public Review):

This study delves into the impact of imidacloprid, an insecticide documented for its toxicity towards honeybees, on the development of bee larvae. The investigation involved exposing bee larvae to various concentrations of imidacloprid, and observing the resultant effects.

The findings of this study revealed that imidacloprid exerted a dose-dependent delay in the development of bee larvae, marked by reductions in body mass, width, and an overall decline in the growth index. Moreover, at elevated concentrations, imidacloprid was observed to impair neural transmission, induce oxidative stress, inflict damage to the gut, and inhibit hormones and genes essential for development. The larvae were found to engage antioxidant defense systems and deploy detoxification mechanisms to mitigate these effects.

However, the manuscript could be significantly enhanced through several improvements. Firstly, the structure of the manuscript warrants refinement to foster coherence and clarity. Additionally, there is a need for careful reevaluation of the concentrations of imidacloprid employed in the study, to ensure their relevance and applicability. In terms of references, greater attention to accuracy in citation is imperative.

Furthermore, while the authors have provided an overview of the general effects of imidacloprid on both vertebrates and invertebrates, the inclusion of a more exhaustive literature review with a specific focus on honey bees and other insects would bolster the context and significance of this research. This would be particularly beneficial in the introduction section, which should be subjected to a major revision.

In summary, this study offers preliminary evidence of the detrimental effects of imidacloprid on the development of bee larvae by interfering with molting and metabolism. This research holds potential as a valuable resource for assessing the risks posed by pesticides to juvenile stages of various animal species.

Reviewer #2 (Public Review):

This study provides evidence on the ability of sublethal imidacloprid doses to affect growth and development of honeybee larva. While checking the effect of doses that do not impact survival or food intake, the authors found changes in the expression of genes related to energy metabolism, antioxidant response, and P450 metabolism. The authors also identified cell death in the alimentary canal, and disturbances in levels of ROS markers, molting hormones, weight and growth ratio. The study strengths come from applying these different approaches to investigate the impacts of imidacloprid exposure. The study weaknesses are not providing an in-depth investigation of the mechanisms behind the impacts observed and not bringing the results in light of the current literature. For instance, the authors' hypothesis is based on two main points, the generation of ROS that leads to gut cell death and energy dysfunction, and the increased P450 expression. They propose this increases P450 expression which in turn increases energy consumption and could contribute to developmental retardation. There is however no investigation on the mechanisms of ROS generation (it could be through mitochondrial damage, Nox/ Duox activity, NOS activity, P450s activity, etc). A link between higher P450 expression and increased energy consumption leading to energy deprivation is also missing. It would also be important for the authors to provide a more complete literature review as previous works have investigated imidacloprid sublethal dose impacts in larval stages for bees and other insect models.